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Ageing mitigation of 21700 Li-ion cylindrical batteries for heavy duty BEVs through forced air and immersion cooling strategies in Nordic climate zones

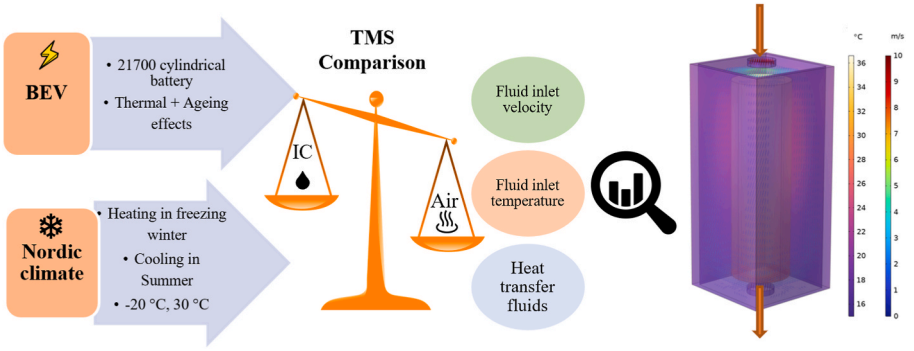
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HIGHLIGHTS

- Effect of Nordic climate conditions on 21700 cylindrical cells is studied.
- High-fidelity physics-based coupled 1-D to 3-D thermal battery model is utilized.
- Thermal and flow design parameters are investigated for air and immersion TMS.
- Heating and cooling capabilities are investigated for both TMSs.
- Parametric sweep includes 234 cases comparing laminar and turbulent flows.

GRAPHICAL ABSTRACT



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ABSTRACT

A Battery Electric Vehicle (BEV) pack requires an intelligent and integrated arrangement of the thermal management system (TMS). While previous studies have investigated TMSs independently, we here make a comparative analysis between air and immersion TMSs tailored for both sub-zero and peak summer temperatures, mimicking Nordic climate zones. A parametric sweep of flow and thermal design parameters using laminar and turbulent flow regimes is performed. These ambient conditions and loading pattern corresponding to a heavy BEV constitute the basis of the comparison. The analysis is based on various well-defined key performance indicators. Moreover, the air based TMS requires high Re flows to give the best performance, while immersion TMS can perform well in a laminar range. It was deduced that low Re (126) and high Re (>3000) may not give better thermal performance, and it is therefore crucial to design the TMS within the proposed flow and thermal design sets. In contrast to previous work, where the main focus has been on the cooling efficiency or steady-state performances, this work extends the horizon to investigate the effect of these TMSs on ageing behaviour to quantify long-term benefits on battery reliability and safety. The coupled ageing-thermal battery model was simulated for 1000 cycles for different cases, showcasing 3.4 % improvement in SOH for immersion TMS

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compared to the air based TMS. Thus, the climate resilient TMS aids in heat management, and the coupled model is thereby shown to help estimate the battery SOH.

Abbreviations

BEV	Battery Electric Vehicle
DF	Di-Electric Fluid
HTC	Heat Transfer Coefficient
HTF	Heat Transfer Fluid
IC	Immersion Cooling
IF	Improvement Factor
lam	Laminar
MRF	Mesh refinement factor
RTR	Rate of Temperature Rise
SEI	Solid Electrolyte Interphase
SOC	State of Charge
SOH	State of Health
TMS	Thermal Management System
turb	Turbulent

Nomenclature

T_{amb}	Ambient temperature (°C)
T_{inlet}	Fluid inlet temperature (°C)
v_{inlet}	Fluid inlet velocity (°C)
Nu no.	Nusselt number
Re	Reynolds number
Pr	Prandtl number
ΔP	Pressure drop (Pa)
d	Diameter (m)
a	Area of inlet (m^2)
ρ	Density (kg/m^3)
Cp	Heat capacity at constant pressure (J/kg.K)
Q_g	Heat generation rate (W/m^3)
u	Velocity Field (m/s)
k	Thermal conductivity (W/m.K)
μ	Viscosity (pa.s)
α	Thermal diffusivity (m^2/s)
Air_3^{30}	Case name for HTF flowing at 3 m/s and inlet temperature 30 °C (example)
$DF_{1\ turb}^{30\text{-}}$	Case name for HTF flowing at 1 m/s (turbulent mode) and inlet temperature 30 °C (example)
$DF_{0.2}^{25\text{-}lam}$	Case name for HTF flowing at 0.2 m/s (laminar mode) and inlet temperature 25 °C (example)
$T_{battery}$	Battery surface temperature (K)
T_{avg}	Battery average temperature (K)
T_{max}	Battery maximum temperature (K)
T_a	Air domain temperature (K)
P	Pressure (pa)
k'	Turbulent kinetic energy (m^2/s^2)
ω	Specific dissipation rate (1/s)
j	Current density (A/m^2)
α_a	Anodic charge transfer coefficient
U_{ref}	Open circuit voltage (V)
R_{film}	SEI resistance ($\Omega \bullet m^2$)
K_{crd}	Volume expansion factor
η_{int}	Overpotential term (V)
ϕ_s	Potential term in solids (V)
ϕ_e	Potential term in electrolytes (V)
r_n	Particle radius (m)
Q_{SEI}	Capacity loss (SEI layer growth) (Ah/m^2)
Q_{Li}	Capacity loss (lithium-plating) (Ah/m^2)
T_{1800s}	Battery temperature at the 1800th second into the driving cycle (°C)
T_{450s}	Battery temperature at the 450th second into the driving cycle (°C)

1. Introduction

Driving range, charging capabilities and battery ageing continue to be crucial questions for Battery Electric Vehicle (BEV) performance [1]. In this context, state-of-health (SOH) is an important but complex parameter which conveys the battery's remaining life. Its depreciation depends on factors such as charge-discharge cycles, temperature, battery chemistry, and usage patterns.

Temperature is a vital operating parameter for lithium-ion (Li-ion) batteries [2]. The optimum temperature range is different for charging (0 °C–45 °C) and discharging conditions (−20 °C–60 °C), as given by cell manufacturers [3,4]. The Li-ion batteries must also be stored at 20 °C–25 °C at lower SOC ranges to retain the capacity for longer durations [5,6]. At sub-zero conditions, the capacity degrades fast due to lithium plating, while power output is limited [7,8]. Lithium plating usually occurs at high SOC, lower temperatures, and high charging rates. Thus, rapid charging is a challenge, especially in Nordic climate zones [9,10]. Furthermore, overheating during charging can cause electrolyte degradation, which can lead to thermal runaway. Operating Li-ion batteries at high temperatures (>25 °C) also accelerates ageing mechanisms such as SEI film growth and loss of active anode material [11]. Therefore, an effective and smartly designed Thermal Management System (TMS) is the backbone for long-range, durable, and safe BEVs [12]. The SOH of the battery pack is thereby strictly monitored by external operating conditions. Table 1 highlights how different operating conditions affect the battery pack.

Active and passive are the two ways that TMSs operate for BEVs. An active TMS relies on an external energy source to operate; this includes forced air and liquid cooling systems, whereas the passive cooling system includes a TMS, which does not require any input energy to operate, such as Phase Change Materials (PCM), fins, etc. Often, the terms Direct and Indirect TMSs are used to specify the modes of contact of the heat transfer fluids. Direct TMS is thus used for immersion cooling systems, and indirect cooling system is used for liquid cooling systems incorporating a channelled plate. Each cooling method has pros and cons, and the implementation requires an in-depth understanding of the cooling or heating requirements. These requirements, in turn, depend on the operating conditions of the EV, which can differ substantially depending both on the climate zone and the user. Air cooled battery packs are adopted by for example Renault ZOE and Nissan Leaf [13]. However, air cooled systems are less reliable for high-temperature ambient conditions or heavy-duty electric vehicles, due to their limited cooling capacity. Air cooling systems are also susceptible to a non-uniform temperature distribution among the cells in the battery pack. Still, air cooled battery packs are employed in many EVs since these are simple in design,

Table 1
Operating conditions affecting the EV battery pack.

Condition	Impact
Temperature	Risk of thermal runaway at high-temperature ambient conditions, capacity degradation at sub-zero conditions (lithium plating). Crucial for life, safety, and performance.
Charge/Discharge Rates	High C-rates can generate excess heat, which can call for parasitic power consumption from the cooling gears. The end results can be reduced EV range and battery pack lifetime.
Customer usage patterns	Battery pack abuse during charging/discharging can affect the battery pack's lifetime and safety. External mechanical stress (vibrations) due to driving style can also damage the battery pack.
State of charge (SOC)	Storing the battery pack at 100 % SOC can reduce the lifespan of the EV battery pack.
Voltage	Both over- and under-voltage can do irreversible damage to the battery pack and can lead to electrolyte decomposition, gas formation, and eventually pave the way for a thermal runaway.
Environmental factors	Conditions such as high humidity can cause corrosion of the terminals and increase the risk of a short circuit.
High altitudes	Slower heat dissipation, an increase in the core and surface cell temperature, and local hot-spots are some of the effects that EVs experience in mountainous regions. The same is applicable to aircraft. This is mainly due to low air pressure (air-density reduction).

cost-effective and are non-toxic without leakage issues. In contrast, liquid-cooled battery packs show better thermal performance owing to their high heat transfer coefficient (HTC) rates [14]. But liquid cooling systems can be complex, and therefore, a balance between parasitic power consumption, thermal performance, and cost needs to be obtained [15]. Indirect TMSs are widely adopted by the majority of EV manufacturers. Passive cooling strategies employ PCM and leverage its high latent heat to store or release the heat. However, these materials do not dump the heat into a sink; rather, they store it. Therefore, they need an additional cooling system to carry away the heat from the system [16]. Fins can help discard the excess heat into the sink in combination with PCMs [17].

Immersion cooling (IC) is currently being explored, as it presents several advantages over other TMSs. IC is achieved by a non-corrosive di-electric fluid (DF). IC systems offer high HTC values and can be used directly without the need of cooling channels, thereby decreasing the contact resistance between the battery and the coolant interface. IC based battery packs can perform better during thermal runaway conditions by preventing fire propagation [18]. However, low thermal conductivity and potential leakage constitute major challenges. The TMS can be applied in single phase or two-phase systems. In single phase, the fluid remains in one state, whereas in dual phase, the fluid undergoes a phase transition from fluid to gas. Two-phase IC systems provide even more enhanced HTC and higher cooling capacity owing to the latent heat of vaporization [19]. IC systems are better capable of lowering the battery temperature as compared to indirect cooling methods [20]. The IC system also has a lower response time to reach a set temperature value, and increasing the DF flow rate decreases the maximum temperature and the temperature difference in the battery [21,22]. A previous study found that the selection of DF and mass flow rate impacts the battery temperature and capacity [23]. However, increasing the flow rate will call for higher pumping power with low improvements in thermal performance [24,25]. IC TMSs generally have high pressure drops due to the high viscosity, which calls for higher pump work than for other TMSs. For example, a battery module immersed in mineral oil with a fluid flow rate of 0.35 m/s was observed to have a pressure drop of 342 Pa due to the viscosity of the DFs [26]. Mass flow rate is thus a crucial parameter that defines the thermal performance of the IC system, and it has been shown that increasing the inlet flow velocity for IC systems does not enhance the thermal performance, but rather increases the parasitic power consumption [24,27–29]. For example, it was observed in one study that when the fluid mass flow rate of the DF was increased from 0.4 L/min to 2 L/min, the decrease in battery's maximum temperature dropped from 4.1 °C to 1.0 °C [30]. Nevertheless, IC systems help in ageing mitigation as they can keep the battery at uniform temperature values; for example, the capacity was improved by >3 % after 600 cycles in one study [31]. IC systems also inhibit the thermal runaway propagation in battery packs and therefore increase the BEV safety [32]. IC systems can also allow a high charging rate of 3.9 C for a 21700 cylindrical cell module by keeping the maximum battery temperature under 30 °C [33]. The Kreisel battery system, which immerses the 18650 and 21700 battery cells in a thermal management fluid, is already installed in commercial vehicles such as the London electric bus fleet [34].

This work presents a comparative analysis between air and IC systems in Nordic zones, using battery requirements adopted for heavy-duty vehicles. An effective thermal management system, especially taking the challenges of Nordic climates into account, needs to quickly heat and cool the battery. A comparative study between air and immersion cooling for data centre application found that the latter can lead to a 50 % reduction in energy consumption and a two-third decrease in spatial requirements [35].

Designing an efficient thermal management system for BEVs, which is not only cost-effective, but also thermally stable, non-corrosive, non-complex, light-weight, and has a quick response rate (i.e., carries the heat effectively to and from the battery pack) remains a technical challenge. A coupled 3-D thermal and 1-D electrochemical model of 21700 NMC Li-ion battery based on the pseudo-two-dimensional (P2D) concept has been utilized to study the effect of thermal regulations on the battery degradation mechanisms. In contrast to previous modelling studies in this area, we here not only compare different TMSs but also compare these under both turbulent and laminar conditions, and we also couple these to a battery ageing model. Moreover, we adapt the situation to the Nordic climate, thereby carrying out the simulations for both cooling (summer) and heating (winter) requirements. The 3-D model was loaded with a discharging cycle corresponding to a heavy BEV. The coupled model renders a clear demonstration of the efficacy of one TMS over the other by comparing the ageing mitigation behaviours of both TMSs.

The work is focused on designing an efficient heat-sink which helps cap the temperature between 20 °C and 40 °C, which is preferred to achieve optimal performance [36,37]. The framing of flow and thermal design parameters is based on a parametric study where thermal performance is assessed for various ambient temperatures ranging from sub-zero levels to Nordic summers against various HTC values. The laminar and turbulent flow design parameters are then swept to find the best candidates. The cooling systems are analyzed based on various key performance indicators (KPIs) that determine the efficacy of the TMS in both cooling and heating requirements during different seasons. The modelled SOH is based on SEI growth and Li-plating. All simulations in this study are performed by making an effort to reduce the simulation time, which reflects an awareness of the carbon footprint of computational research methods [38].

2. Model development

This work has utilized a coupled 1D-to-3D battery model for studying the effect of the proposed TMSs on the degradation mechanisms. The coupled electro-thermal model takes temperature as input from the 3-D thermal model, and the 3-D thermal model takes the heat generation from the 1-D electrochemical model. For the selection of the best heat-sink design, only the 3-D thermal battery model has been employed to lower the complexity. The modelling has been carried out in COMSOL 6.3.

The popular cylindrical cell Li-ion battery type 21700 NMC532 (Nickel-Manganese-Cobalt; $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$) has been considered in this study. Both cell types, 18650 and 21700, can be efficiently cooled from the base or the sides [39]. A comparative study between prismatic and cylindrical battery types concluded that cylindrical battery types can store more heat than prismatic battery types, even for low ambient conditions [40]. Moreover, thermal management of 21700 cylindrical cells is more challenging than the smaller 18650 format due to higher ratios of volume-to-surface and active-to-non-active material ratios [41].

2.1. 3-D thermal model

Two popular working fluids were investigated for heating and cooling applications in active mode: (i) air and (ii) dielectric fluids (DF). A comparative analysis was carried out for winter (−20 °C) and summer (30 °C) periods, and the best cases in terms of performance according to the KPIs (section 3) were selected for the life-time prediction model. The flow type, i.e., turbulent or laminar, was selected based on the parametric study to effectively heat/cool the battery. Table 2 shows the thermo-physical properties of the 3-D model. The model consists of a cylindrical battery and an acrylic box with an inlet and outlet at the base and top of the box. The approach to utilize an acrylic box has been employed previously to isolate the battery pack and serve as a control volume (fluid-domain) [42,43]. The terminals of the cylindrical battery were insulated. It is important to note that such acrylic boxes are only utilized experimentally for research purposes, where they serve as a transparent control volume, which renders it possible to view the

Table 2
Thermophysical properties [44].

Property	Battery	Air	Silicon oil	Acrylic-box[a]
Density (kg/m^3)	2867	1.26	918	1190
Thermal conductivity (W/m.K)	Vertical direction = 33.9, parallel direction = 0.8	0.026	0.167	0.18
Heat capacity (J/kg.K)	830	1005	1620	1470
Viscosity (Pa.s)	–	1.8e^{-5}	5.1e^{-3}	–
Prandtl number, Pr	–	0.71	49.5	–
Thermal diffusivity, α (m^2/s)	–	0.2e^{-6}	8.67e^{-6}	–

[a] from COMSOL materials database.

thermal development by an infrared camera. The model here aims to study the flow characteristics of the two HTFs and the corresponding effect on thermal and ageing behaviour. Thus, the cell-to-cell thermal interaction is not included in the model. Fig. 1 shows the schematic of the model, and Table 3 gives the geometrical specifications of the 3-D model.

The heat transfer for solid and fluid domains is given by Eq. (1):

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p u \cdot \nabla T + \nabla \cdot q = Q_g \tag{1}$$

$$q_c = -k\nabla \mathrm{T} \tag{2}$$

Where T is the temperature (K), ρ (kg/m^3) is the solid density, C_p (J/kg·K) is the solid heat capacity at constant pressure, k (W/m·K) is the solid thermal conductivity, u (m/s) is the velocity field, and Q_g (W/m^3) is the heat source. The heat conduction (q_c) is defined by Fourier's law of heat conduction, defined by Eq. (2).

The heat flux leaving/entering the battery for cooling/heating application is given by Eq. (3) as per Newton's law of cooling/heating. The heat q_0 denotes the normal heat flux (W/m^2) added or lost to the boundary due to convection:

$$q_0 = HTC\ (T_{battery} - T_a) \tag{3}$$

where HTC is the heat transfer coefficient, $T_{battery}$ is the battery surface temperature, and T_a is the temperature of the air domain. The surfaces where thermal insulation condition is applied are defined by Eq. (4).

$$n.\ q_0 = 0 \tag{4}$$

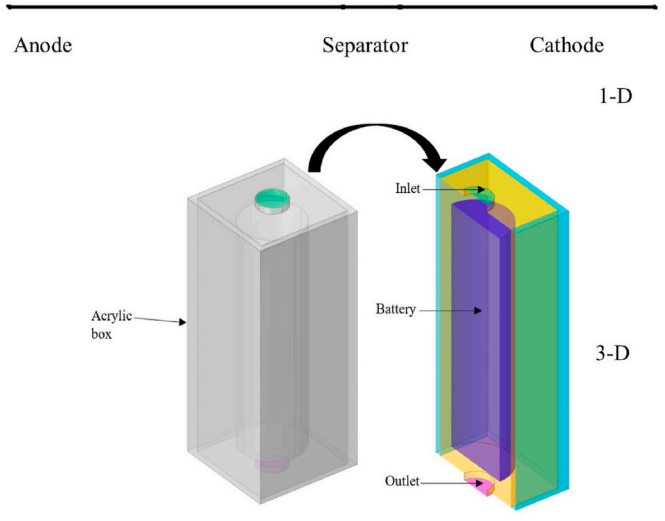


Fig. 1. Schematic of the 1-D and 3-D thermal battery model.

Table 3
Geometrical specifications [44].

Specifications	Value
Battery type	21700 NMC Li-ion cylindrical battery
Anode	Graphite
Cathode	$Li(Ni_{0.5}Co_{0.2}Mn_{0.3}O_2)$
Capacity	5 Ah
Max-charge voltage	4.2 V
Cut-off Voltage	2.5 V
Internal resistance	0.015 Ω
Height	70 mm
Diameter	21 mm
Weight	69.5 g
Acrylic box	35 x 80 × 2 mm^3
Inlet/outlet	Diameter 10 mm, thickness 2 mm

where n is the normal vector to the boundary.

The fluid enters at fluid inlet temperature T_{inlet}. At the inlet boundary in the fluid domain, a boundary condition is applied that accounts for the heat flux that would be carried by the incoming fluid. This is accomplished by applying Danckwerts condition on the enthalpy, which expresses the normal conductive heat flux (q) at the inlet as a function of fluid density and velocity (u) (flow rate) and the variation in enthalpy (ΔH) across the fluid domain from inlet to outlet:

$$n.q = \rho \Delta H u.n \tag{5}$$

$$\Delta H = \int_{T_i nlet}^{T1} C_P\ dT \tag{6}$$

This study utilizes both turbulent and laminar flow models in COMSOL. The flow is incompressible. RANS-EVM, k-ω model has been utilized to model the turbulent flow for both fluids. The dependent variables are velocity field components (u, v, and w, m/s), pressure (P, Pa), turbulent kinetic energy (k', m^2/s^2) and specific dissipation rate (ω, 1/s). No slip condition was applied to the walls. The fluid enters at the inlet as a fully developed flow. The equations that define the physics of the problem are governed by momentum, continuity, and energy equations (turbulent and laminar), and are defined in Appendix A.

2.2. 1-D model and battery degradation mechanisms

A physics-based pseudo-two-dimensional (P2D) electrochemical model is developed to study the degradation mechanisms of 21700 NMC-based lithium-ion batteries (LiBs). This model, being part of a locally developed digital twin framework [45], is enhanced to capture the coupled effects of SEI layer growth and lithium plating on the anode particles, coupled with thermal effects [46]. Appendix C, Table C.1 lists the battery model parameters.

2.2.1. Main intercalation reaction

The primary lithium-ion intercalation or deintercalation reaction occurs at the surface of the active material particles and is governed by the Butler-Volmer equation:

$$j_{int} = j_{int,0}\left[exp\left(\frac{\alpha_a F}{RT}\eta_{int}\right) - exp\left(\frac{(1-\alpha_a)F}{RT}\eta_{int}\right)\right] \tag{7}$$

where $j_{int,0}$ represents the exchange current density, α_a is the anodic charge transfer coefficient, and η_{int} is the overpotential associated with lithium-ion intercalation or de-intercalation, defined as:

$$\eta_{int} = \phi_s - \phi_e - U_{ref} - j_{tot}R_{film} \tag{8}$$

where ϕ_s and ϕ_e denote the potentials in the solid and electrolyte phases, respectively. U_{ref} is the open circuit voltage, and R_{film} is the resistance of the SEI layer on the graphite particles.

When side reactions occur at the negative electrode interface, their

contributions are incorporated into the overall reaction kinetics. Accordingly, the total current density j_{tot} is expressed as the sum of the intercalation current j_{int}, the SEI formation current j_{SEI}, and the lithium plating current j_{lpt}: $j_{tot} = j_{int} + j_{SEI} + j_{lpt}$

2.2.2. SEI layer growth

The formation of SEI layer initiates during the first charge cycle. The initial thickness of its inner inorganic and outer organic layers are specified as follows:

$$l_0^{inorganic} = \frac{\delta Q_1 M_{SEI}^{inorganic}}{2A_0 \rho_{SEI}^{inorganic} F} \qquad (9)$$

$$l_0^{organic} = \frac{(1-\delta) Q_1 M_{SEI}^{organic}}{2A_0 \rho_{SEI}^{organic} F}$$

where A_0 denotes the ideal initial surface area of the anode particles:

$$A_0 = 4\pi R_n^2 + 8\pi R_n^2 \rho_{cr,0} l_{cr,0} a_0 \qquad (10)$$

The SEI layer resides at the interface between the anode and the electrolyte. During charging, the expansion of graphite particles can cause mechanical stress, leading to cracks in the microporous SEI layer. These fractures expose fresh surfaces, promoting additional SEI layer growth [47]. As a result, the current density of the SEI side reaction is given by:

$$j_{SEI} = -\left[\left[1 + HK_{crd}\left(\frac{C_{n,av}}{C_{n,max}}\right)\right] \cdot \frac{J j_{1c,loc} C_{rate}}{exp\left(\frac{\alpha_a F}{RT}\eta_{SEI}\right) + \frac{Q_{SEI} f J}{j_{1c,loc}}}\right] \qquad (11)$$

where J, H, and f are lumped parameters representing the dimensionless exchange current, the relative expansion factor of graphite, and the frequency, respectively [47]. The term K_{crd} is the volume expansion factor related to the stoichiometric ratio $C_{n,av}/C_{n,max}$ in Li_xC_6. The overpotential of the SEI layer, η_{SEI}, is defined as $\eta_{SEI} = \phi_s - \phi_e - U_{eq,SEI}$, where the equilibrium potential $U_{eq,SEI}$ is set to 0.4 V. $J_{1c,loc}$ refers to the local current density based on a 1C charge/discharge rate:

$$j_{1c,loc} = \frac{Q_0}{A_v \cdot L_n} \qquad (12)$$

where Q_0 is calculated as $Q_0 = C_{p,max}(x_{p,max} - x_{p,min})\varepsilon_{s,p} L_p F$. The specific surface area, A_v, is given by $A_v = 3\varepsilon_n / r_n$. ε_n denotes the volume fraction of active material in the negative electrode, and r_n is the radius of its particles.

The concentration of the SEI layer, C_{SEI}, is determined based on mass conservation, expressed as $\frac{\partial C_{SEI}}{\partial t} = -\frac{A_v j_{SEI}}{2F}$. The capacity loss due to SEI layer growth, denoted Q_{SEI} (Ah/m^2), is given by:

$$Q_{SEI} = \frac{\int_{negative} C_{SEI} \cdot F}{3600} \qquad (13)$$

2.2.3. Lithium plating and stripping

Lithium plating and stripping are represented as side reactions governed by a concentration-dependent Butler–Volmer formulation.

$$i_{lpt} = i_{lpt,0}\left(\frac{C_l}{C_{l,ref}}\right)^{0.5}\left[F_a\, exp\left(\frac{F\eta_{lpt}^n}{2RT}\right) - exp\left(-\frac{F\eta_{lpt}^n}{2RT}\right)\right] \qquad (14)$$

where $j_{lpl,0}$ denotes the lithium plating current density at the anode particle interface, given as $j_{lpl,0} = k_{lpl} c_l^{\alpha_{lpl}}$. The pre-exponential factor for the anodic term, F_a, is defined as:

$$F_a = \frac{c_{Li(s)}}{c_{Li(s),cov}}, \frac{c_{Li(s)}}{c_{Li(s),cov}} \le 1$$
$$F_a = 1, \frac{c_{Li(s)}}{c_{Li(s),cov}} > 1 \qquad (15)$$

The overpotential triggering the lithium plating reaction, η_{lpl}, is given by:

$$\eta_{lpl} = \phi_s - \phi_e - j_{tot} R_{film} \qquad (16)$$

The concentration of deposited lithium, comprising both reversible metallic lithium ($C_{lpt,\,metal}$) and dead lithium ($C_{lpt,dead}$), is calculated using mass conservation as follows:

$$\frac{\partial C_{lpt,metal}}{\partial t} = -\frac{V_{metal} A_v j_{lpt}}{F}$$
$$\frac{\partial C_{lpt,dead}}{\partial t} = -\frac{V_{dead} A_v j_{lpt}}{F} \qquad (17)$$

The stoichiometric coefficients are defined as V_{metal}, representing reversible lithium during plating and stripping, and V_{dead}, representing the dead lithium fraction during stripping. They are set as follows:

$$V_{metal} = 1, V_{dead} = 0, when\ \eta_{lpl} \le 0$$
$$V_{metal} = 1 + Y_{dead}, V_{dead} = -Y_{dead}, when\ \eta_{lpl} > 0 \qquad (18)$$

where Y_{dead} represents the stoichiometric coefficient for the dead lithium fraction. The corresponding capacity loss due to lithium plating and stripping, $Q_{Li,dead}$ (Ah/m^2), is given by:

$$Q_{Li,dead} = \frac{\int_{negative} C_{lpt,dead} \cdot F}{3600} \qquad (19)$$

2.3. Validation

The coupled electrochemical-thermal model, based on the mathematical model described in section 2.2, is sourced from previous work and is validated at charge/discharge conditions 1.3 C and 15 °C [48]. The properties for the 21700 NMC532 battery type are sourced from the COMSOL Multiphysics 6.3 materials database and a study by Yin et al. [41]. As shown in Fig. 2a, the coupled model closely matches the experimental temperature curve, demonstrating its ability to represent real battery operating conditions accurately [48]. This shows that the mathematical coupled model is well-suited for accurately simulating real-time battery scenarios. However, this previous work was based on a similar battery type (18650-NMC532), and therefore, the coupled model for the 21700 NMC532 battery type was also validated against experimental work by Yang et al. [49,50]. The battery was discharged at 1 C, 2 C, 3 C, 4 C, and 5 C rates. The discharge cut-off voltage was set as 2.5 V, and the peak voltage was set to 4.2 V. The ambient temperature was maintained at 25 °C. The validation (Fig. 2c) reveals that the predicted cell surface temperature for various discharge rates agrees with the experimental values with errors of 0.99 %, 1.67 %, 0.23 %, 0.6 %, and 0.41 % respectively, for 1 C–5 C discharge rates at the end of the discharging cycle.

The battery format used by Yang et al. is INR-21700-P42A (manufactured by MOLICEL). The manufacturing differences modify the parameters of the battery model, impacting voltage drops. The cell capacity of INR-21700-P42A is 4.2 Ah, whereas in this study, the cell capacity is 5Ah, which impacts the voltage drop. Therefore, the cell capacity was changed to 4.2 Ah, and the voltage drop trend was compared with the data presented in the data sheet by MOLICEL [51]. Fig. 2f shows the impact of cell capacity on voltage drop and validates the model at 4.2 Ah cell capacity. Additionally, the datasheet was also used to validate the cell capacity degradation at 10 A and 20 A currents for 500 cycles. The charging was done using CC-CV protocol at 4.2 A till the voltage became 4.2 V and then keeping it fixed till the cut-off current of 50 mA is reached, and the discharging was done at 10 A and 20 A

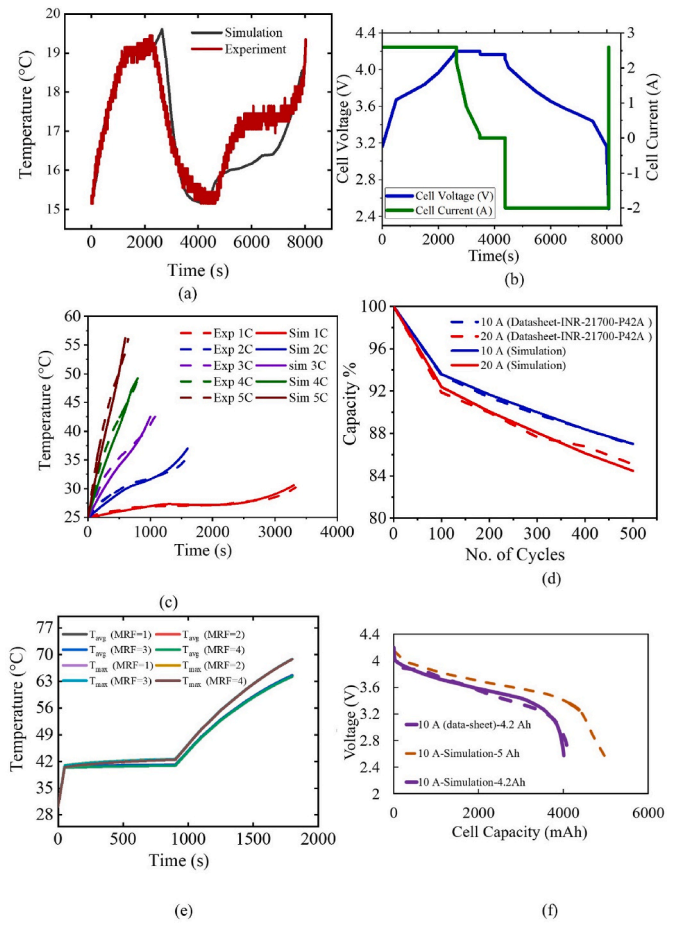


Fig. 2. (a) Comparison of surface temperature profiles between simulation and experimental results during cycling at 1.3 C and 15 °C, (b) Current-voltage profile used for validation, (c) 21700 battery temperature validation for 1 C–5 C discharge rates against experiment work, (d) Capacity degradation comparison with the datasheet for INR-21700-P42A (e) Mesh dependence study (f) Voltage comparison at different cell capacity against INR-21700-P42A MOLICEL-datasheet.

(CC), with 10 min rest after 10 A and 20 min rest after 20 A. Fig. 2d shows that the data sheet data matches the simulation data, which gives the ageing prediction further validation. At 10 A discharge, the Root Mean Square Error (RMSE) between simulation and datasheet capacity fade is 0.17 %. At 20 A discharge, the RMSE is 0.50 %, indicating that the aging model can effectively capture the degradation behavior of the 21700 cells. Thus, these studies show the robustness of the coupled 1-D to 3-D electrochemical-thermal battery model in terms of thermal and ageing predictions.

2.4. Mesh independence study

A mesh-independence study was performed to verify that the results obtained are consistent for different mesh elements. The mesh refinement factor (MRF) was defined, and the lower values denote a lower number of elements in the mesh (coarser mesh). Fig. 2e shows that the results are stable, and no significant variation was observed. Thus, the mesh with MRF 1 was selected for all other simulations.

3. Key performance indicators (KPIs)

The performance analysis is depicted by the heating and cooling performance, the temperature uniformity, and the energy consumption. Therefore, the following KPIs have been selected.

The thermal performance of the two cooling systems is assessed based on the following KPIs:

1. Rate of temperature rise (RTR): The measure of how quickly the battery can be heated [52].

$$RTR = \frac{T_{min} - T_{initial}}{t} \tag{20}$$

2. Improvement factor (IF): The measure of how quickly the battery can get cooled.

$$IF = \frac{T_{avg.before_{cooling}} - T_{avg.after_{cooling}}}{T_{avg.before_{cooling}}} \tag{21}$$

3. Temperature gradient (ΔT): ΔT is an important design criterion for TMSs [53]. Therefore, monitoring the thermal gradient during cell operation is critical and should be less than 5 °C [54].

$$\Delta T = T_{max} - T_{min} \tag{22}$$

4. Heat flux ($\dot{q}$): $\dot{q}$ is the amount of heat absorbed by the TMS. The heat removal rate is defined by Eq. (3).

5. Pumping power (W_P): W_P depends on density, flow rate, and pressure drop of the HTF. The variation of W_P decides the energy required to pump the fluid to keep the TMS running and, therefore, is a decisive factor when any type of TMS is chosen. This is an important factor in designing an efficient TMS, since it gives rise to parasitic power consumption.

$$6.\ W_P = \Delta P.a.v_{inlet} \tag{23}$$

7. Cooling/heating power (W_C): The work required to maintain the T_{inlet} greatly affects the efficiency of the TMS. The coefficient of performance (COP) is assumed to be constant and is set at the value 2 [55]. The analysis is also further conducted for the energy consumed in summer and winter per kilometre (Wh/km) for the given drive cycle.

W_C is given by the equation:

$$W_C = \frac{\dot{m}\ .\ C_P\ .\ (T_{ambient} - T_{inlet})}{COP} \tag{24}$$

$$E_{km} = \frac{W_c\ (W).\ \text{drive cycle time (h)}}{\text{drive cycle length (km)}} \tag{25}$$

8. Reynolds number (Re): The laminar and turbulent flows for air and DF are based on the Re given by Eq. (26). Re no. is a KPI as it shows: (i) how the flow rate and HTFs affects the thermal performance and (ii) how different flow profiles (laminar and turbulent) affect the thermal performance.

$$Re = \frac{\rho.u.d}{\mu} \tag{26}$$

9. Heat transfer coefficient (HTC): The HTC is dependent on the Nusselt number (*Nu*), diameter (d), and thermal conductivity (k) of the fluid. The value of *Nu* can be calculated from Eq. (28) according to the Dittus-Boelter correlation [56]. HTC has a direct impact on battery cooling and thus is a KPI, as also explained in Eq. (3).

$$HTC = \frac{Nu.d}{k} \tag{27}$$

$$Nu = 0.023\ \left(Re^{0.8}\right)\left(Pr^{n}\right) \tag{28}$$

$$n = \begin{cases} 0.4 \text{ for heating} \\ 0.3 \text{ for cooling} \end{cases}$$

4. Results and discussion

4.1. Case without heating and cooling

The operation mode in this study mimics heavy-duty BEVs in Nordic climatic zones. Table 4 highlights the power values for different vehicle operations designed for a fully-electric 490 kW truck, which has been scaled down to be adapted for a single 21700 cylindrical cell [57]. The Scaling factor depends on the capacity of a single cell and the truck pack. The scaling factor is the ratio of the energy Wh in a single cell and the entire truck pack. The scaling factor can be utilized for scaled-down power calculations, which can return the values of current for different driving modes. In the real packs, there can be thousands of such 21700 cylindrical cells, and there are modules that are representative of a single energy unit, which are then cooled together by either air or cooling plates. The drive cycle consisted of acceleration for 1 min, followed by highway driving (cruise mode) for 16 min, and then ending with an uphill operation for 13 min. The drive cycle thus lasts for 30 min, covering 40 km.

Fig. 3a and b show the heat generated and the current v/s time graph against the drive cycle [58,59]. These simulations were carried out without any TMS, under natural convection. Fig. 3c shows the variation of volume average temperature with time for winter and summer without TMS. The initial temperature for reference was set as 25 °C for summer and −20 °C for winter. The volume average temperature at the end of the cycle (T_{1800s}) was 80.7 °C and 36.8 °C, respectively, for summer and winter. It was also noted that if the initial temperature in summer was set in equilibrium with the surroundings, the battery temperature after the drive cycle was 85.6 °C, which defines the worst-case scenario. During the initial period of the drive cycle, the temperature is in the sub-zero range till 850 s. This calls for heating the battery before the start of the operation. On the other hand, during the summers, the temperature rises sharply, and a cooling strategy is thus required.

4.2. Laminar vs. turbulent flows

TMS design for Nordic climates brings challenges, which pertain to the fact that there is a huge difference in T_{amb} during winter and summer seasons. This study compares the IC strategy against the common air-cooling strategy in forced convection mode. The 21700 cylindrical battery was checked for laminar and turbulent HTC values. Fig. 3d shows the variance of the battery's thermal performance (i.e., battery temperature at the end of the load cycle) with T_{amb} and HTC values. T_{amb} was varied between −20 °C and 30 °C, and HTC was varied from 10 to 300 W/m^2.K, simulating all combinations. This parametric study, which comprised 90 cases, revealed that higher values of HTC will have better thermal performance in Nordic climate conditions. Also, a threshold is found for the HTC value (200 W/m^2.K) beyond which there is little to no improvement in the thermal performance (Appendix F). The HTC value also depends on the Re, which in turn is dependent on the flow channel's spatial properties as well as thermo-physical properties. The Re decides whether the flow is laminar or turbulent, being laminar below Re 2000 [60]. Thus, the Re was varied from 2000 to 7000. For example, for air h = 51.4 W/m^2.K (axial-fans), u = 10 m/s, Re = 5515, i.e., Re > 2000, which indicates modelling of turbulent conditions.

Table 4
Operation mode based on heavy duty BEV.

Driving-mode	Power (kW)	Current (A)	Scaled-down power for single 21700 cell operation (W)
Acceleration at 110 km/h (1 min)	400	500	14.4
Cruise at 90 km/h, highway (16 min)	200	250	7.2
Climbing uphill at 65 km/h (13 min)	350	437	25.2

The working fluids were air and silicon oil (used as DF). The velocities were varied such that the Re was kept within the specified limit for comparative analysis. The comparative analysis helps uncover the important details in the thermal and flow parameters, like temperature difference, which arises due to the velocity gradient that influences the HTC on the battery surface, the effect of fluid inlet temperature for different HTFs, etc., which cannot be captured through the HTC values when assessing the TMSs. The cases were simulated for three T_{inlet} values, i.e., 15 °C, 25 °C, and 30 °C, for both summer and winter conditions. Table 5 shows the parametric sweep for the 3-D flow model in COMSOL. The 8 different scenarios for air-based TMS and 8 different scenarios for DF-based TMS have all been simulated for three different inlet temperatures and two different ambient temperatures, thereby rendering 96 cases to be simulated for the comparative study between air and IC TMSs. The IC system was also investigated for laminar flow, and Table 6 lists the details of those 48 cases (8 scenarios with three different inlet temperatures and two ambient temperatures) that were examined. All simulated cases have been organised and labelled in Table D.1 (appendix) to facilitate the analysis of the results; the cases are named so that the temperature is indicated as superscript and the v_{inlet} as subscript. Re depends on the size of the inlet, the material properties of the heat transfer fluid (density, viscosity), and the fluid inlet velocity. It is important to note here that the Re values were varied according to the fluid inlet velocities.

To assess the thermal performance of the two cooling methods, the battery was kept in equilibrium with the ambient temperature initially, thereby modelling the worst-case scenario. The heating is assessed for a battery at −20 °C and how fast the TMS takes to bring the battery temperature above 5 °C. The same case is for cooling in summer, where the ambient temperature is 30 °C. The initial temperature for reference was set as 25 °C for summer and −20 °C for winter. The volume average temperature at the end of the cycle (T_{1800s}) was 80.7 °C and 36.8 °C, respectively, for summer and winter. It was also noted that if the initial temperature in summer was set in equilibrium with the surroundings, the battery temperature after the drive cycle was 85.6 °C, which defines the worst-case scenario.

4.3. Parametric sweep: air as working fluid

Fig. 4a shows the temperature and velocity field contours for the case with T_{inlet} 15 °C, v_{inlet} 10 m/s and T_{amb} 25 °C, simulated for 1800 s (30 min) with the turbulent flow model using air as the working fluid (also see Figure B.1 for an enlarged view). This figure also highlights the changing HTC value on the battery surface with constant v_{inlet}. This is a common phenomenon in EV battery packs where not all cells can be cooled or heated at the same HTC. Fig. 4b highlights how v_{inlet} and T_{inlet} contribute to the thermal performance of the battery for summer. A lower v_{inlet} with lower T_{inlet} seems to provide satisfactory thermal performance. However, a higher fluid velocity with a medium range fluid inlet temperature can show promising thermal behaviour with low parasitic power consumption. Thus, a v_{inlet} above 8 m/s for air cooled battery packs will show better cooling performance in the summer.

Winter time calls for heating, and Fig. 4c–d show the variation of thermal performance against T_{inlet} and v_{inlet}. The hot air enters the inlet at the start of the drive cycle and therefore shortens the duration that the battery is exposed to the negative temperature. Fig. 4c shows that the temperature for the winter condition is above 0 °C for all the data sets after 450 s. However, it is also noted that the temperature increment is faster for higher Re values than for lower Re values. Fig. 4d shows the temperature at the end of the drive cycle for the winter condition. It is observed from Fig. 4b–d that a v_{inlet} above 8 m/s and a $T_{inlet} \geq 15$ °C can

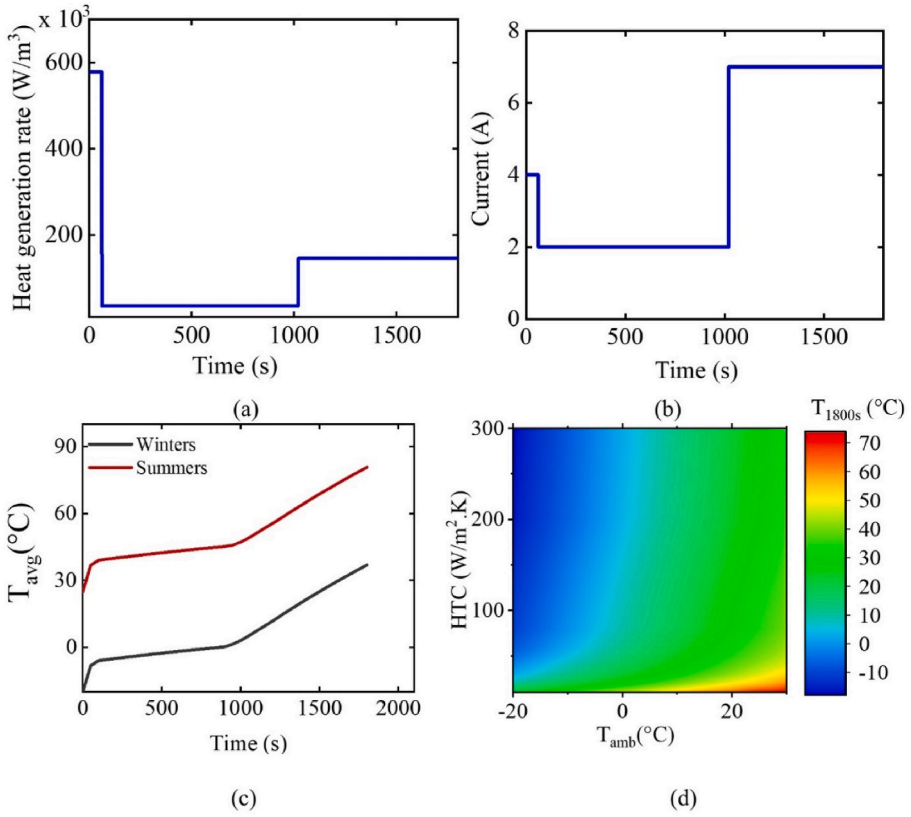


Fig. 3. Scaled down heavy EV load profiles for the cylindrical battery (a) heat generation rate (b) current, (c) temperature vs. time plot (without heating/cooling) for a battery utilized at −20 °C T_{amb} (winter) and 30 °C T_{amb} (summer), (d) Impact of HTC on the surface temperature at the end of the load cycle.

Table 5
Parametric sweep of the thermal-flow model for comparative analysis of air- and DF-based TMSs.

v_{inlet} (m/s) (Air)	Re (Air)	v_{inlet} (m/s) (DF-turb)	Re (DF-turb)
3	2088	1	1800
4	2785	1.2	2160
5	3481	1.5	2700
6	4177	2	3600
7	4873	2.5	4500
8	5569	3	5400
9	6265	3.5	6300
10	6961	4	7200

Table 6
Parametric sweep for laminar flow model with DF as working fluid.

v_{inlet} (m/s)	Re (DF-laminar)
0.8	1500
0.6	1000
0.5	900
0.4	700
0.3	500
0.2	360
0.1	180
0.07	126

work well for both seasons by maintaining a temperature ≤45 °C. This observation is crucial as it simplifies the thermal management system design when one criterion fits both seasons. The results also reveal that turbulent air flows can decrease the requirement for parasitic power consumption, especially for heating and cooling of the HTF, as high velocities can be combined with high fluid inlet temperature in the summer time, which can help in reducing the energy expenditure. Fig. 4e–f show the importance of high Re (high velocities) for air cooling systems. The variation of volume average temperature with time based on the input drive load is shown for different fluid velocity values for T_{inlet} = 25 °C. The Re values are listed in Table 6 for each velocity value. The volume average temperature at the end of the drive cycle for Re = 2088, T_{inlet} = 25 °C, is 59.9 °C, and for Re = 6961, T_{inlet} = 25 °C, is 43.9 °C. The temperature for Re = 6961 thereby dropped by 36.8 °C in comparison to the case without any TMS for summer conditions.

4.4. Parametric sweep: DF as working fluid

The DF cooling was simulated for laminar and turbulent flows. The turbulent flow can keep the battery temperature at a constant value of 25 °C in summer, which outperforms the air-cooling system for similar Re flows (Fig. 5a). However, the energy expenditure of turbulent DF flows necessitates repeating the simulations for the IC system's laminar situations. The DF was then simulated for laminar flow conditions, and the Re was varied from 1500 to 126 (Table 6). A comparative analysis of laminar and turbulent flow conditions in Fig. 5b, (c) and 5(d) highlights the effectiveness of laminar flow over the turbulent flow for the IC system. Fig. 5b–d show the poor performance zones for v_{inlet} 0.07 m/s, 0.1 m/s and 4 m/s for summer and winter conditions, respectively. The laminar flow case with Re = 126 and v_{inlet} = 0.07 m/s shows poor performance in cooling the battery (Fig. 5b). Also, for v_{inlet} 0.07 m/s, 0.1 m/s and 4 m/s, it is observed from Fig. 5d that these cases took longer to heat the battery than other cases. These cases also showed unsatisfactory thermal performance for summer conditions, as shown in Fig. 5b. Thus, in forced IC systems, the flow condition affects the thermal performance, and the Re must be equal to or above 360 for effective cooling. The same is valid for heating applications during winter, where the flows at 0.07 m/s, 0.1 m/s, and 4 m/s take longer to bring the

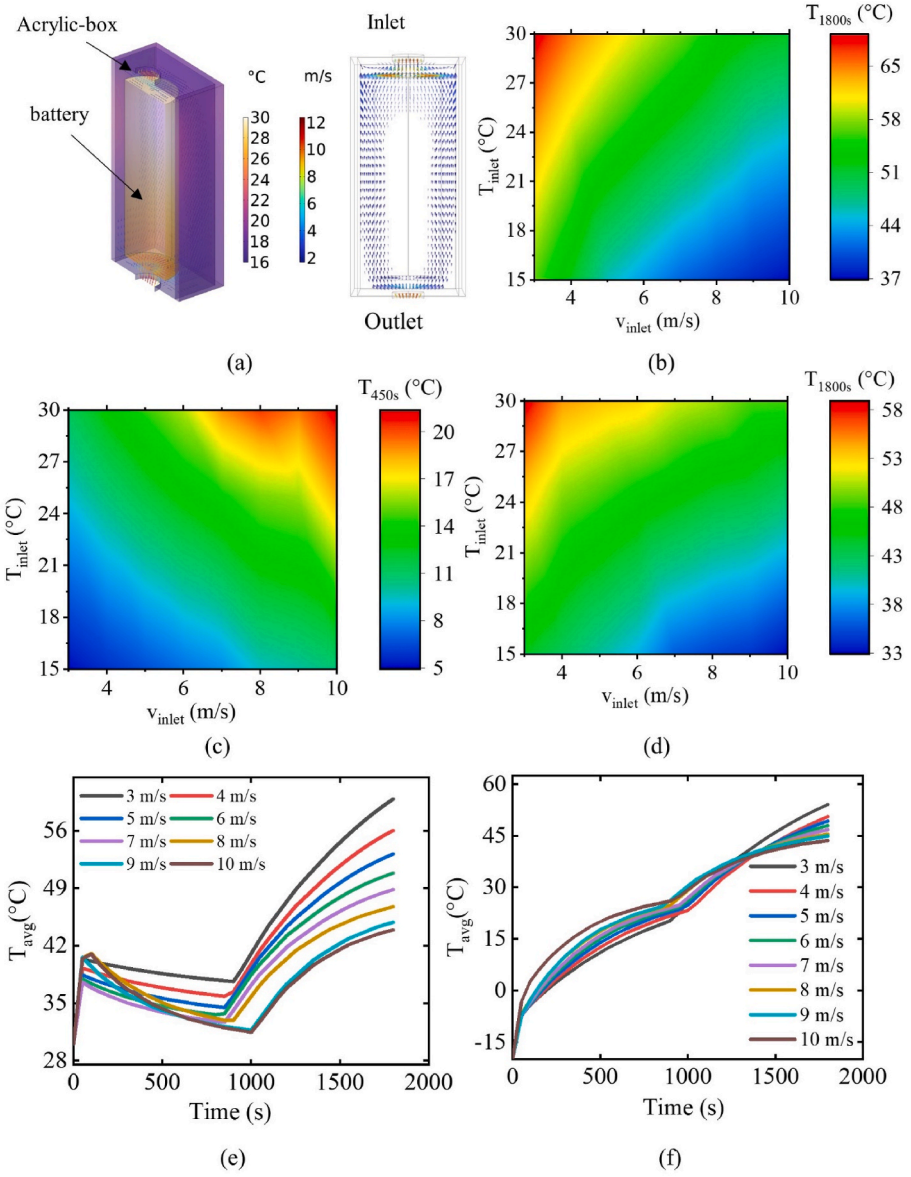


Fig. 4. (a) Temperature and velocity field contours around the battery at T_{inlet} 15 °C, v_{inlet} 10 m/s, and T_{amb} 25 °C with air as working fluid, (b) Effect of v_{inlet} and T_{inlet} on the battery temperature when cooling is required, (c) Battery thermal performance when heating is required at 450 s, (d) Effect of v_{inlet} and T_{inlet} on the battery temperature when heating is required at 1800s, Volume average temperature plot for T_{inlet} = 25 °C during (e) summer, (f) winter.

battery's temperature to above 0 °C (Fig. 5c–d). Therefore, increasing the v_{inlet} only has a limited enhancement in the thermal performance. The turbulent flows (Re > 3000), in turn, exhibit poor thermal performance. The possible reason could be the contact time, which is small when fully utilizing the heat transfer ability of the IC system [28]. Other reasons are geometrical constraints due to the positions of the inlet and outlet (non-uniform distribution of velocity on the cylindrical battery surface), flow by-passing, and limited thermal conductivity of DF.

4.5. Comparative analysis between air and DF cooling and heating for the various KPIs

The comparative analysis between air and DF is based on the KPIs mentioned in section 3.

i Rate of temperature rise (RTR)

RTR is an important KPI when heating is required. The rate of temperature rise during the initial phase of the driving cycle impacts the battery's capacity to perform optimally. The heating in this study is based on the assumption that the battery pack is at T_{amb}, which encompasses extreme temperatures in the Nordic climate. This assumption provides a worst-case scenario against which the thermal performance of the two proposed heating methods is examined. The best heating should, in principle, happen for the case in which the fluid moves faster and enters at a relatively higher temperature at the inlet. In this study, this case corresponds to case no. Air_{10}^{30} for the air heated battery. The time at which the battery's temperature is above 5 °C is set as a reference for a comparative investigation. The time taken to heat the battery above 5 °C is 150 s for this case. The RTR for the case Air_{10}^{30} is 0.18, which is a non-dimensional value. The DF, in contrast, maintains a constant temperature across the driving cycle, and the RTR is also higher than for air. The RTR for case $DF_{1}^{30-turb}$ is 0.96, while the RTR for the case $DF_{0.07}^{30-lam}$ is 0.76. It is observed that RTR for air, even at high Re (6961), can only provide values around 0.18, whereas RTR with DF can achieve values as high as 0.76 even for low Re (126). Thus, IC based TMS can serve as a better choice for Nordic climate zones as it can heat the battery quickly from sub-zero temperatures.

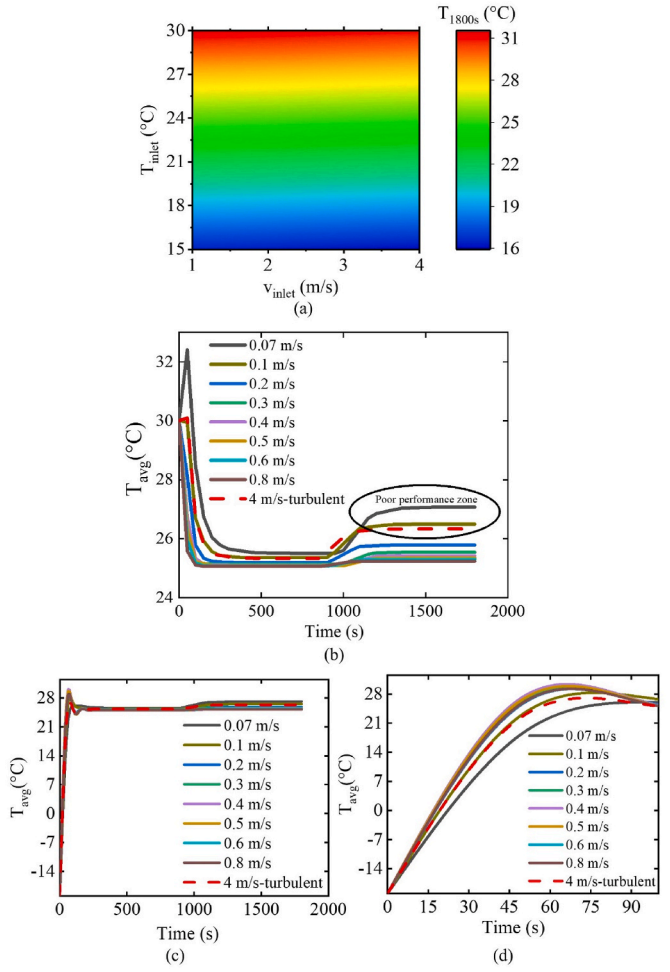


Fig. 5. (a) Effect of v_{inlet} and T_{inlet} on the battery thermal performance at 1800s when cooling is required (turbulent flow case), Comparative analysis of laminar flow and turbulent flow for DF at T_{inlet} = 25 °C, (b) summer (c) winter (d) enlarged view, winter.

ii Improvement factor (IF)

IF defines the effectiveness of the cooling system. Fig. 6a-c shows that the IC system is more efficient in cooling the battery than the air cooling system in summer conditions. As can be seen, the maximum value of IF for air cooling was 0.6. On the other hand, the maximum values of IF for IC were 0.81 and 0.82, respectively, for laminar and turbulent flow cases. The results obtained clearly show that DF is better HTF than air with 37 % enhancement in the IF.

iii Pumping power

Pumping power has a direct dependency on the viscosity and flow rate of the fluid. The pressure drop for DF (laminar), having v_{inlet} = 0.8 m/s, can be seen in Fig. 6d-f, and is 5.8 times more than for air at 10 m/s. These values are in agreement with some previous studies [24,28]. The viscosity of the DF is considerably higher than that of air, and the pumping power is therefore also required to be higher. This is an important factor when designing an efficient TMS, since it gives rise to parasitic power consumption, which also motivates to find a suitable laminar flow regime at lower fluid inlet velocities, which can offer reduced energy input requirement.

iv. Heating (W_h) and cooling power (W_C)

The heating/cooling power required to operate the DF-based TMS is higher than for the air based system. The DF is denser than air, and this will therefore add to the parasitic power losses. The choice of the working fluid is therefore a critical part of the TMS, which is correlated to the Re value. For example, the W_c required for case Air_{10}^{25} (Re = 6961) is 2.45 W, and for case $DF_{0.5}^{25-lam}$ (Re = 900) is 143.58 W, and the W_h required for the same cases are 22.01 W and 1292.25 W, respectively. It is also observed that the cooling and heating requirements are much lower for DF in laminar flow conditions in comparison to turbulent flow, which is important to consider to minimize the power losses.

v Heat transfer coefficient (HTC)

HTC is an important parameter when designing effective heat sinks, a it signifies the amount of heat being carried away by the moving fluid at the battery surface. However, this alone cannot represent the total scenario of heating or cooling. The local values of velocity magnitude across the battery vary as per the flow conditions. It was observed that for the cases Air_{10}^{25} and $DF_4^{25-turb}$, the maximum magnitude of velocity in the fluid domain is only 1.67 m/s and 0.67 m/s, respectively. This appears because of the battery's position with respect to the inlet. The fluid flow is in the z direction, which slows down as the HTF travels from top to bottom till the outlet. Therefore, constant values of the HTC are unrealistic in these thermal simulations. The cells placed in the battery pack are also prone to such variation in HTC, which also gives rise to unequal cooling or heating. The calculated value of HTC for case Air_3^{30} from the Dittus-Boelter correlation (Eq. (27)) is 24.4 W/m^2.K, while the value from the COMSOL simulation is 8.9 W/m^2.K. These deviations show that the correlations can only give estimated values, but the real heating or cooling depends on the design of the heat sink and the properties of the HTF. The amount of heating or cooling done depends on various other factors, as is shown in this study. The v_{inlet}, fluid inlet temperature, and thermal properties greatly impact the heating or cooling, together with the heat sink design.

vi Temperature difference (ΔT)

ΔT must be kept within the permissible limits (<5 °C). Fig. 7a-d demonstrates the variation of ΔT for various cases. The thermal performance depends on how the values of maximum and minimum temperature vary for different case numbers. While the DF offers a homogenised temperature range at the end of the drive-cycle, air displays a non-uniform temperature distribution. It is also evident from Fig. 7a-b that at higher values of velocity, air can perform satisfactorily by keeping the ΔT around 6 to 5, but the IC system offers better thermal performance for this KPI.

vii Reynolds number (Re)

Re depends on the HTF properties, v_{inlet}, and the diameter of the inlet. The similar Re for air and DF results in different thermal performances. Fig. 7e-h shows a comparative analysis for both summer and winter condition. The temperature at the end of the discharge cycle for summer and winter is 43.9 °C and 43.6 °C, respectively, for the air cooling system. The DF with the same Re performs better than air by keeping the average temperature at 26 °C in both summer and winter conditions.

viii Heat flux ($\dot{q}$)

Heat flux is an indicator of the effectiveness of the heat-sink design in most thermal management system applications. If investigated meticulously, this factor reveals how well the battery pack can be thermally regulated in various phases of the charge/discharge or driving cycles. It was observed from Fig. 7i that the heat flux value (W) at low velocities for the air-cooled battery is positive, which indicates that the TMS is

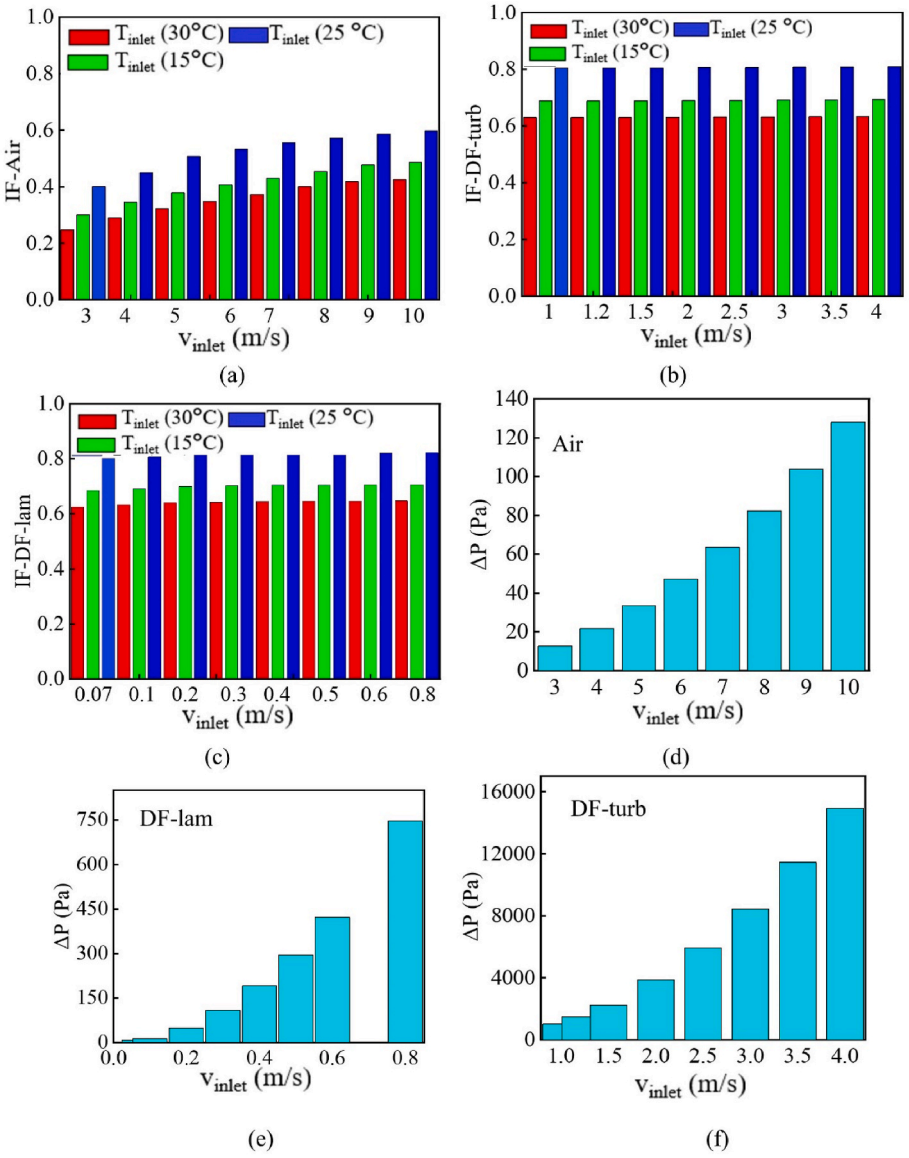


Fig. 6. Comparison based on the pumping power requirements using (a) air, (b) DF-turbulent, (c) DF-lam cooling, and pressure drop for cases with HTF (d) Air, (e) DF in laminar zone, (f) DF in turbulent zone.

inefficient towards the end of the driving cycle. This means the heat in the acrylic box is accumulated over time and is not being carried away by the HTF. This phenomenon is also observed for the IC system for high T_{inlet} values in turbulent flow simulations. The heat flux improves with increasing v_{inlet}, but the less contact time limits the thermal performance, as explained in section 4.4. It is also observed that with low values of v_{inlet} (laminar cases, Fig. 7j), the DF performs satisfactorily with heat flux values < 0, unlike air, where the heat flux value is > 0 for low values of v_{inlet}. Appendix G, Figure G.1 explains the DF-turbulent heat flux map and its thermal performance in terms of heat flux against DF-laminar cases.

In summary, these results highlight the various factors that determine the effectiveness of using DF over air as HTF. The Nordic climate zones can be a challenge for the smooth functioning of BEVs. The parametric studies show that in both cooling and heating scenarios, DF can keep the battery within a constant temperature range with a temperature difference <1 °C. It also highlights the need for high Re (>3000) for air for high thermal performance. On the other hand, medium ranged Re (200-1500) can work well for DF. The IF points towards the efficacy of the IC system over the air cooling system when cooling is required. On the other hand, the comparative analysis also revealed that high thermal performance with DFs comes with high pump work, but DF can perform well under laminar flow. Moreover, the improved thermal performance will result in enhanced safety for the IC system by maintaining an ideal temperature of 25 °C in both seasons.

5. Impact on ageing and energy

5.1. Ageing analysis

The coupled electrochemical-thermal model was simulated to analyse the impact of the TMSs on battery ageing over 1000 cycles of charging/discharging. The discharging cycle was kept the same, as described in Section 4.1, and the battery was discharged until its cut-off voltage of 2.5 V. The battery was charged at a 1C rate via a CC-CV protocol in which it was first charged till 4.2 V, and then kept at a fixed voltage until reaching the cut-off current 0.25 A. The two best cases for air and DF were simulated for ageing analysis and compared against the case without TMS. Thus, the cases simulated were (i) Without TMS, at 30 °C ambient temperature, (ii) Air[25], summer, and (iii)

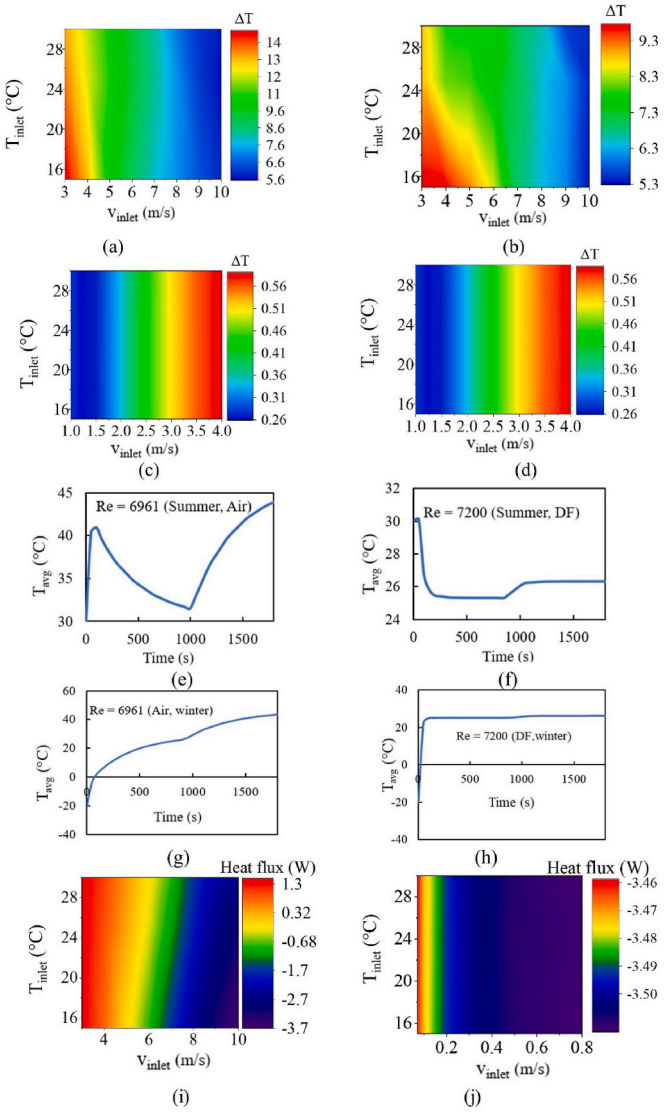


Fig. 7. Temperature difference at T_{1800s} (a) Air (summers), (b) air (winters), (c) DF (summers), (d) DF (winters); Temperature v/s time plot for the case (e) Air, $v_{inlet} = 10$ m/s, $T_{inlet} = 25$ °C, $T_{amb} = 30$ °C, (f) DF, $v_{inlet} = 4$ m/s, $T_{inlet} = 25$ °C, $T_{amb} = 30$ °C, (g) Air, $v_{inlet} = 10$ m/s, $T_{inlet} = 25$(°C), $T_{amb} = -20$ °C, (h) DF, $v_{inlet} = 4$ m/s, $T_{inlet} = 25$ °C, $T_{amb} = -20$ °C; (i) Heat flux map for air cooling, (j) Heat flux map for IC system for laminar flow cases.

$DF_{0.5}^{25-lam}$, summer. Fig. 8 shows the SEI growth, Li plating thickness, and SOH for these cases.

The SEI thickness impacts the resulting battery surface temperature by directly affecting the electrical resistance, heat generation, and ion transport efficiency. It is therefore essential to track its growth over the cycle numbers [61]. Fig. 8a shows the growth of the SEI layer and Li-plating thickness over the life cycle simulations. The obtained values agree with previous studies [62–67]. The results show that the SEI growth for air and IC, respectively, was reduced by 44.16 % and 61.6 % from the case without any TMS. The Li-plating for the cases without any TMS, Air, and DF at the end of 1000 cycles were 4.98 nm, 15.16 nm, and 27.66 nm, respectively. The results showed that the Li plating thickness increased with a decrease in temperature, and SEI growth increased with a rise in battery temperature. It was also noted that the average battery temperature over the discharge cycle was 56.71 °C, 38.8 °C, and 25 °C, respectively, for the cases without any TMS, Air, and DF, which justifies the trend of Li plating and SEI growth over the number of cycles [62]. To minimize ageing, the battery temperature should not vary between the

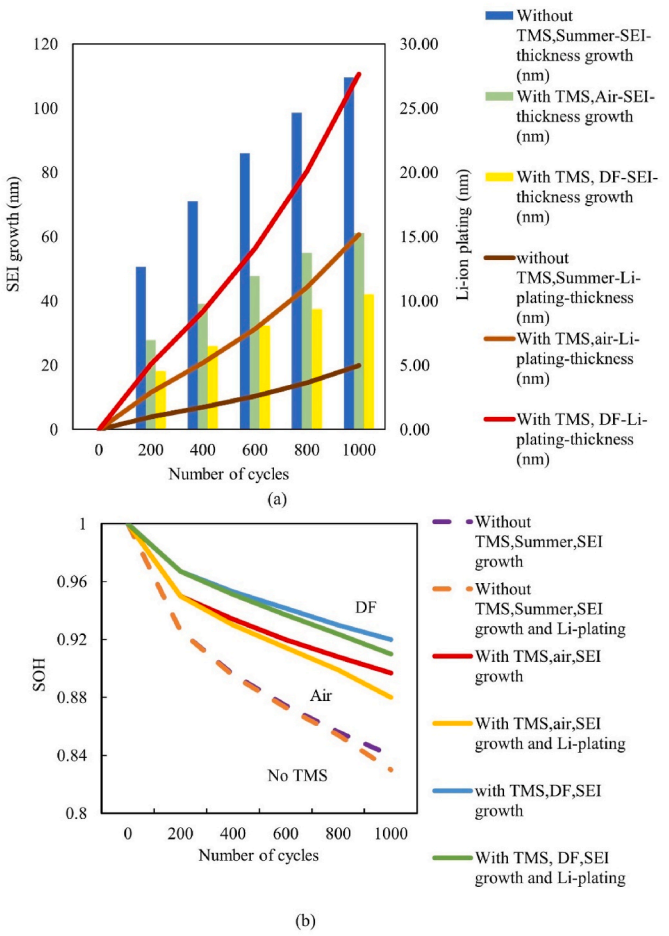


Fig. 8. (a) SEI and Li-plating growth over the cycles; (b) SOH as influenced by SEI layer growth, and by SEI with Li-plating thickness growth together.

two seasons, and this becomes especially relevant for heavy BEVs carrying cargoes long distances from north to south, and vice versa. More energy will then be required to heat the cargo, powertrain, and battery pack. The battery SOH shown in Fig. 8b, affected by SEI growth and Li-plating, was improved for DF by 9.63 % and 3.4 % respectively, from the cases without TMS and air. Fig. 8b shows how the battery's SOH is affected by different cooling/heating strategies.

5.2. Energy analysis

This section discusses the energy demand requirement (Wh/km) for air and DF, calculated for the driving cycle. Table 7 lists the values of the E_{km} for the best cases, which are compared for winter and summer conditions for the air and IC system and is defined by Eq. (25) in Section

Table 7
E_{km} for different cases.

Scenario	Case attributes	Cycle-Average temperature	E_{km} (Wh/km)
Summer condition without TMS	No TMS	56.71 °C	–
Air_8^{25}, summer	Air flow: 8 m/s, T_{inlet}: 25 °C	38.8 °C	0.025
Air_8^{25}, winter	Air flow: 8 m/s, T_{inlet}: 25 °C	25.07 °C	0.22
$DF_{0.5}^{25-lam}$, summer	DF flow: 0.5 m/s, T_{inlet}: 25 °C	25.35 °C	1.8
$DF_{0.5}^{25-lam}$, winter	DF flow: 0.5 m/s, T_{inlet}: 25 °C	24 °C	16.15

3. Air (Air_8^{25}) has 98.6 % lower energy requirement per km than DF for both summer and winter seasons. Also, the energy requirement for winter conditions is higher compared to summer for both the TMSs. However, air is not efficient in lowering the battery's average temperature during summer. This calls for a better energy system that can effectively cool down the battery. Therefore, DF is recommended for summer conditions. It is also to be noted here that during winter, air can outperform the DF by maintaining a cycle average battery temperature of 25.07 °C for the case Air_8^{25} with a lower E_{km} value of 0.22 Wh/km. The SOH benefit is thus balanced off by the E_{km} requirement, climatic conditions, and driving cycle. Figure E.1 (appendix) shows the variation of temperature for the case Air_8^{25} and $DF_{0.5}^{25-lam}$ for both summer and winter. DF helps maintain a uniform temperature variation of the battery across the drive cycle length.

Apart from ageing and energy parameters, TMS selection for EV battery packs is also governed by thermal performance, cost, ease of installation, and safety. A study by Roe et al. highlighted that immersion fluids can increase the heat transfer by 10,000 times compared to air cooling [68]. Kim et al. further supported this statement by highlighting the fact that IC systems can provide a high HTC even for small inlets with a small hydraulic diameter, despite the high pressure loss, which also supports the results achieved in this work [69]. DFs are generally flame retardants and can keep the cells at a uniform temperature, thereby decreasing the cell-to-cell thermal variation in the battery pack. Therefore, IC systems must be selected for the applications that require efficient cooling, better safety, and large battery packs generating high volumetric heat, such as for heavy BEVs. However, cost, parasitic energy expenditure, and leakage issues are some of the limiting factors that should be taken care of in future studies by the optimum design of the battery pack with an IC system.

6. Conclusions

This study has investigated the importance of TMSs for BEVs, targeting Nordic climates. It could be envisioned that the absence of extreme ambient heat in summer times for Nordic zones may motivate the selection of a traditional air cooling system over other advanced and complex approaches. However, the analysis reveals the efficacy of the IC system in offering better control over battery temperature and ageing mitigation despite its higher energy demands. One of the key observations is that a constant value for the HTC will not capture the interplay of various physical and operational parameters in the TMS model. Instead, there is a strong interdependency of HTC on flow design, material properties of HTFs, and inlet and outlet position, which needs to be taken into account.

The results have been compared for both the cooling and heating capacity of the HTFs using parametric sweeps. While the air cooling system performs significantly better with higher Re values, the IC system shows a limiting trend without further improvements in thermal performance for higher Re (>3000) and lower values of the Re (128). Generally, the IC system performs better than air for the same Re flows. Therefore, a close range of Re (200–1500) values is proposed for the IC system. Thus, this study encourages the selection of advanced climate specific TMS designs that give better BEV system performance along with ageing mitigation of the battery pack. Certainly, the economic and environmental aspects need to be considered for the final choice of the TMS.

In the future, this work can be extended to module and pack level based on the position and number of inlets and outlets for the IC systems. The optimization of material, thermal, and geometrical properties can be helpful for optimized design of the battery pack, where this work can be helpful for providing useful insights.

CRediT authorship contribution statement

Ashima Verma: Writing – original draft, Investigation, Conceptualization. **Wendi Guo:** Writing – original draft, Methodology, Investigation, Conceptualization. **Litao Yin:** Writing – review & editing, Methodology. **Niladri Roy Chowdhury:** Writing – review & editing, Methodology. **Amin Moosavi:** Writing – review & editing, Methodology. **Daniel Brandell:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declares no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2025.238643.

Data availability

Data will be made available on request.

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